

Naqib Sad Pathan

PhD Candidate · Optimal Transport, Computational Biology, Single Cell Data Analysis

✉ qpb3vt@virginia.edu ☎ +1 4347605312
📍 Charlottesville, VA 🌐 [Website](#)
🐙 [GitHub](#) 🔗 [LinkedIn](#)
🏆 [Scholar](#)

ABOUT ME

Doctoral Researcher in Electrical Engineering, University of Virginia (Expected Graduation: Spring 2027) Researcher in the Imaging and Data Science Laboratory under Prof. Gustavo K. Rohde, developing methods at the intersection of optimal transport, machine learning, mathematical modeling, and computational biology. Research focuses on generalizable frameworks for analyzing high-dimensional cellular and subcellular measurements, with applications to flow cytometry, multiplex imaging (4i/cIF), digital pathology, and spatial omics data analysis.

EXPERIENCE

Graduate Research Assistant August 2022 – Present
Imaging and Data Science Laboratory (Rohde Lab), University of Virginia, Charlottesville, VA

Trainee August 2023 – Present
The Systems Analysis of Stress-adapted Cancer Organelles (SASCO) Center, University of Virginia

Research Intern — Spatial Proteomics / Computational Biology Summer 2026
Genentech, South San Francisco, CA

Lecturer / Assistant Professor April 2016 – August 2022
Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh

SKILLS

Languages: Python, C/C++, MATLAB, JavaScript / TypeScript, Bash, Git

Machine Learning & AI: PyTorch, Scikit-learn, High-Dimensional Statistics, Deep Learning, Pointnet++, DGCNN, Deepsets, Optimal Transport, Large Language Model(LLM)

AI Engineering & Developer Tools: OpenCode, Aider, Ollama, Local LLM Deployment, Agentic AI Workflows, Prompt Engineering, VS Code, Antigravity, Jupyter Notebook

HPC & Systems: Linux, Slurm, UVA Rivanna HPC, GPU Computing, Parallel & Distributed Computing, Remote Development, Batch Processing, Environment Management, Docker, SSH

Imaging and Biomedical Data Analysis: Flow Cytometry, Multiplex Imaging (4i/cIF), Digital Pathology, Spatial Transcriptomics, Proteomics, PPG, EEG, Brain Image, Image Processing

REFERENCES

Gustavo Kunde Rohde Advisor
Professor at BME and ECE · University of Virginia
gustavo@virginia.edu · Website: <https://www.imagedatascience.com/>
✉ gustavo@virginia.edu

Kevin Janes Research Core, SASCO
Professor, Biomedical Engineering · University of Virginia
kjanes@virginia.edu · Website: <https://systemsbioe.org/>
✉ kjanes@virginia.edu

Todd Stukenberg Research Core, SASCO
Professor, Biochemistry and Molecular Genetics · University of Virginia
pts7h@virginia.edu · Website: <https://toddstuke.wixsite.com/stukelab>
✉ pts7h@virginia.edu

EDUCATION

PhD in Electrical Engineering 2022 – Expected Spring 2027
University of Virginia, Charlottesville, VA

Dissertation: “A calculus for high dimensional single-cell measurements” · Advisor: Prof. Gustavo K. Rohde · CGPA: 3.71

MSc in Electrical and Electronic Engineering 2016 – 2021
Chittagong University of Engineering and Technology (CUET)
CGPA: 3.71

BSc in Electrical and Electronic Engineering 2010 – 2015
Bangladesh University of Engineering and Technology (BUET)
CGPA: 3.81

PROJECTS

FlowLOT: Linearized Optimal Transport for Flow Cytometry Analysis

Developed FlowLOT, an optimal transport-based framework that represents patient-level flow cytometry measurements as discrete probability distributions and maps them into compact, fixed-length embeddings. The framework enables interpretable analysis of high-dimensional single-cell data, supporting classification, visualization, and continuous outcome prediction within a unified mathematical framework. Demonstrated its utility for measurable residual disease (MRD) estimation in acute myeloid leukemia (AML), achieving transparent and biologically interpretable patient-level modeling.

Optimal Transport, Linear Optimal Transport, Flow Cytometry, MRD Estimation, Computational Hematology, AML, Python, HPC, Deepsets, Pointnet++, DGCNN

📄 [Code](#) 📄 [Paper](#)

Metaphase Chromosome Analysis with Iterative Indirect Immunofluorescence Imaging (4i)

Investigates how chromosome passenger complex (CPC) signaling and chromosome condensation are coordinated during mitosis using high-dimensional multiplex imaging. Develops optimal transport-based methods to quantify, visualize, and compare chromosome-scale variation across cell lines, studying how chromosome geometry influences protein localization and chromosomal organization.

4i Imaging, Multiplex Imaging, Spatial Biology, Chromosome Biology, Chromosome Passenger Complex, Python, Image Processing

Quantifying Nuclear Structures in Digital Pathology Using Transport-Based Morphometry (TBM)

Develops transport-based morphometry (TBM) methods grounded in optimal transport theory to quantify nuclear chromatin organization directly from microscopy images. By representing nuclei relative to a common template, these methods provide interpretable and robust morphological descriptors that reveal cancer-associated structural variation across tissue types and imaging protocols.

Digital Pathology, Whole Slide Imaging, Transport-Based Morphometry, Optimal Transport, Nuclear Morphometry, Python

📄 [Paper](#)

Computational Autofluorescence Correction for Cyclic Immunofluorescence (Genentech Summer Internship Project)

Designed and implemented scalable computational imaging workflows for autofluorescence correction, artifact removal, and image restoration in high-plex cIF experiments. Leveraged Zarr-based storage, Dask, and parallel/distributed computing to analyze terabyte-scale whole-slide and TMA proteomics and spatial transcriptomics datasets generated on the Lunaphore COMET platform.

Multiplex Imaging, Spatial Proteomics, Single-Cell Analysis, Python, Cyclic-IF, Dask, Zarr